Week 4 web page: delivery guide

Instructor notes for Week4_ProbabilityDistributions.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week4_ProbabilityDistributions.html. Open it before class; the three widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week4_ProbabilityDistributions.R` (same seeds, same numbers). Everything runs in a second or two.
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker 4.4-4.6 (printed pp. 155-188); factorials in Appendix A.5 (p. 482).
- WODs: Monday reused *A Probability Puzzle* (Week 3's WOD, key in `wod/answer_keys/`); Wednesday *Expectation and the Sample Mean* (key in `wod/answer_keys/`).

As taught (Sept. 14-16, 2026)

- **Monday:** the probability puzzle WOD (not reached the previous Wednesday), then combinatorics, the birthday problem, random variables, pmf vs. pdf, and expectation and variance, following the PDF NOTES (now in `lecture_notes/archive/`). Per the announcement (added Nov. 4): Monday got through combinatorics (with the WOD), the birthday problem worked together, and the start of random variables and distributions (weights summing to 1, binomial and Poisson domains, why $P(X = x) = 0$ for continuous variables). Probability densities were deferred to Wednesday, so expectation and variance likely were too. The announcement's framing of random variables and distributions is now in `#pmf-pdf`; its "Chapter 5 is the gateway to likelihood" line is in the Week 5 page's `#simulation` summary.
- **Tuesday:** final project pitches; no lab Acts and no lab lecture.
- **Wednesday:** the sample-mean WOD, then the bestiary of distributions (taught from the distributions section of legacy `LECTURE2`; there was no PDF NOTES file). The distributions ran through the whole period, so synthetic data simulation moved to Week 5 Monday. Per the announcement (added Nov. 4), Wednesday also covered probability densities ($f(x)$ as a limit; $f(x)\,dx$ as the continuous analog of $P(X = x)$; now in `#pmf-pdf`) and expectation and variance (the WOD's focus). The 3Blue1Brown Gaussian integral link is now in `#continuous`.

- The page was built Oct. 29, after the week, from those sources plus Bolker 4.4-4.6 and a draft bestiary plan. New on the page: the three widgets, the bear-genotype discussion question, the log-scale bonus (idea bank), method of moments, and the gamma-Poisson bonus.
- `LECTURE2` is archived: its first half (axioms, urns, Bayes' rule, Monty Hall) is on the Week 3 page, built Oct. 30.

Part 1: Counting, random variables, and expectation (Monday, ~35 min)

Big picture (~2 min)

- Point at MODELS. Weeks 2-3 built the deterministic half; this week is the noise half.

WOD (~8 min; runs long)

- Dice grid on the page: 6 sevens, 6 doubles, both 1/6.
- "With two eight-sided dice, which sum is most likely, and does it tie, beat, or lose to doubles?" *Land:* 9 (8 ways of 64), and it ties doubles (also 8 ways). With n-sided dice the most common sum, $n + 1$, always ties doubles: both have n cells.
- Lottery: 142,506 tickets. Ask who forgot to divide by 5! before going on.

Counting (~8 min)

- Ponds: 11,880 ordered visits vs. 495 sets of 4. The ratio is $4! = 24$.
- Nest figure: 15 arrangements of 2 failures in 6 nests, each with probability $0.25^2 \times 0.75^4$; total $0.297 =$ `dbinom(2, 6, 0.25)`.

Birthday problem (~7 min)

- Ask for a show of hands on the WOD guess first. 25 people: 0.569; 300 pairs.
- Widget, in order: $n = 25$ (0.569), slide down to 23 (0.507, 253 pairs) and 22 (below 0.5). "Fill 100 groups" a few times at 25 to show the simulated rate near 0.57.
- "With 365 values, find the smallest group with better than even odds of a match. How many pairs does it have? Switch to 10,000 values and find the smallest such group again. What do the two pair counts have in common?" *Land:* 23 people (253 pairs) and 118 people (6,903 pairs). Both pair counts are about 0.69 times the number of values ($\ln 2 \approx 0.69$): matches become likely once the number of pairs approaches the number of possible values. Group size grows only like the square root of the number of values.

pmf vs. pdf (~8 min)

- Bolker's pH example: a reading of 7.9 means 7.82-7.98. Uniform on [7.0, 7.1] has density 10.
- Trout: $P(20 \leq X \leq 24) = 0.383$; the density 0.0997 at 22 cm is not a probability.

- Widget: start at the default (SD 4, width 4: height 0.100, height × width 0.398, area 0.383). Then SD down to 0.10 (height 3.99, area still at most 1). Then width down to about 0.01 (area and height × width agree and both shrink toward 0).
- "Shrink the SD to about 0.1 cm. The curve now rises above a height of 1. Is the model broken? Then, at any SD, narrow the interval width toward zero and watch the 'Area' and 'Height × width' cards. What happens to each, and what does that say about the probability of one exact length?" *Land:* not broken; height is probability per cm, and only areas must stay at or below 1. As the width shrinks, area ≈ height × width, and both go to 0: any single exact length has probability 0. With a wide interval at SD 0.1, height × width can exceed 1, which is why it only works for narrow intervals.

Expectation and variance (~8 min)

- 40 routes: sample mean 1.775 vs. $E[X] = 1.5$. Use this to separate "computed from data" from "property of the distribution" (sets up Wednesday's WOD).
- **Board:** derive $\text{Var}(X) = E[X^2] - \mu^2$ (expand the square, apply linearity). All three versions give 1.125.
- Mode 1, median 1, mean 1.5 for the nest distribution.
- "The expected number of failed nests is 1.5, but no survey can record 1.5 failures. In what sense is 1.5 'expected'?" *Land:* it's the long-run average over many surveys (the balance point of the distribution), not a value any one survey is expected to produce. The most likely single outcome is the mode, 1.

Discussion (~3 min)

- Q1 (pH density 19.9) *Land:* nothing is wrong. The density is per pH unit; a reading within 0.02 of 7.9 has probability 0.68, an area. A narrow distribution has to be tall to enclose an area of 1.
- Q2 (150 bears, chance match 1 in 10,000) *Land:* 11,175 pairs, so $P(\text{at least one shared ID}) \approx 1 - e^{-1.12} = 0.67$. Very worried: two bears become one "individual," so the population estimate is biased low (the "shadow effect"). Fix: genotype more loci to shrink the chance-match probability; at 1 in a million the probability drops to about 0.011.

Bonus: log scale (skip if behind)

- `factorial(365)` is `Inf`; `lfactorial()` gives the same 0.431. Mention that likelihoods will need the same trick.

Part 2: Tuesday

- Final project pitches; nothing from the page.

Part 3: Probability distributions (Wednesday, ~35 min)

WOD (~8 min; runs long)

- Parameter vs. statistic, then $\text{Var}(\bar{X}) = \sigma^2/n$; the SE is $\sigma/\sqrt{n}$ (students mix these up). Simulation: variance 3.27 vs. 3.2, SD 1.81 vs. 1.79. The three histograms narrow from $n = 1$ to $n = 20$.
- "The variance formula assumed the five trout were independent. Suppose all five came from one pool, where fish share food and temperature and so resemble each other. Would the variance of $\bar{x}$ be larger or smaller than $\sigma^2/5$? Why?" *Land:* larger. Fish that resemble each other move together, so their errors don't cancel as much; five similar fish carry less information than five independent ones (a smaller effective sample size). This is the WOD's 2b condition; covariance itself comes later.

Four functions (~5 min)

- Density at 22 cm 0.0997; 2.3% longer than 30 cm; 90th percentile 27.13 cm. `pbinom(2, 6, 0.25)` = 0.831 includes 2; $P(X \geq 3) = 0.169$.

Discrete (~7 min)

- Poisson(4) vs. NegBin(4, $k = 1.5$): variance 4 vs. 14.67; empty quadrats 0.018 vs. 0.142 (7.8 times as many).
- Poisson as the limit of the binomial: one sentence; students saw the numerical comparison in Lab 3.

Continuous (~6 min)

- Bolker's gamma is shape and *scale*; R's default second argument is rate. Name the lognormal and point to the Week 5 page.
- Widget, in order: Poisson at $\lambda = 1$ and 20 (ratio stays 1); Negative binomial at mean 4, k from 0.5 (ratio 9) to 20 (ratio 1.2); "Draw 200 values" once to compare sample and true variance.
- "Choose Poisson and move λ from 1 to 20 while watching the 'Variance / mean' card. Then choose Negative binomial, set the mean to 4, and move k from 0.5 to 20. Which distribution could describe counts whose variance is 3.5 times their mean, and roughly what k would it need?" *Land:* only the negative binomial; its ratio is $1 + \mu/k$, so $k = 4/2.5 = 1.6$ (the card reads 3.50). The Poisson is stuck at 1.

Choosing (~6 min)

- Seed counts: mean 4.2, variance 14.9, ratio 3.55; method of moments $\hat{k} = 1.65$. The Poisson gives the 16-seed quadrat about a 7 in a million chance.

Discussion (~3 min)

- Q1 (clutch size 2-5, Poisson mean 3.4) *Land:* the Poisson puts 15% of its probability on clutches of 0 or 1 and 13% on more than 5, all impossible. Real clutch sizes are also far less variable than a Poisson (variance well below the mean, underdispersion); check the variance-to-mean ratio. Options: model clutch size minus 2 as a binomial with $N = 3$, or treat it as categorical.

- Q2 (seed variance 3.5 times the mean) *Land:* dispersal limitation (seeds land near parents, so they arrive in clumps) and heterogeneity among quadrats (microsites, distance to a parent, soil). In a model: let the rate depend on a covariate such as distance to the nearest parent, or let it vary randomly among quadrats (gamma-Poisson, the bonus).
- Q3 (beta vs. binomial for leaf damage) *Land:* binomial needs countable trials with a known total, such as N grid squares on each leaf scored eaten or not, each with the same probability and independent (damage is clumped, so even that would be overdispersed). Beta fits a continuous fraction from image analysis, but can't produce exact 0s (undamaged leaves) or 1s, which need a mixture.

Bonus: gamma-Poisson (skip if behind)

- Simulated mean 3.95 and variance 14.29 (theory 4 and 14.67). k is the shape of the among-quadrat rate distribution; random effects pick this idea up later.